SPARTAN MVP PDD · ATLAS SERIES · MAY 2026

ARK Platform MVP PDD — SPARTAN Compressed

SCM 7-Step compression of ARK Platform PDD v3 (12 surfaces · 89 routes · 34 tables) into a single-developer, 4-week, Replit-conducted MVP — 24 prompts, 5 phases, $CR_p \approx 73\%$.

> **We don't build AI agents.** We engineer the DNA that governs them — powered by your cognition, owned by you. The MVP ships exactly the loop that proves that claim.

ATLAS PROMPTWARE SERIES
JNGL-ARK-MVP-2026-001
ATANDA STUDIO · SPARTAN SI V1.0.0

ATANDA
STUDIO

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Phase 5 · Billing · Privacy · Launch (Week 4 late)

Part 4 · Upgrade Path

1. Compression Header

```
Compression ID:   JNGL-ARK-MVP-2026-001
Source PDD:       JNGL-ARK-PDD-V3-2026 (12 surfaces · 89 routes · 34
tables · 88 source prompts)
MVP PDD:          JNGL-ARK-MVP-2026-001 (24 MVP prompts · 4 weeks)
Compressor:       SPARTAN SI v1.0.0  ·  SPRT-ATLAS-MVP-SI-2026-001
                  JCSE 49/50  ·  □ Wolf
Co-Agent:         ADA ULTRA SI  ·  Seat #1 (Architecture Authority)
                  TARANTULA  ·  Seat #2 (Replit Build Discipline)
ZPOS+5:           PRISM (P0 prompts · 96.8%) ·
                  QUANTUM (route/SSE technical · 95.4%) ·
                  SYNTHESIS (worksheet rows · 96.7%)
Target VIBE App:  Replit Agent (92/100 · PRIMARY)
                  v0.dev (84/100 · session musician for component pre-
gen)
Date:             May 2026
Status:           □ SPARTAN COMPRESSION COMPLETE · CERTIFIED · ATLAS MVP
DELIVERED
```

> *"ARK PDD v3 documents twelve surfaces, every one of them strategically real. The MVP keeps the seven that close the loop — Identity, Resume, CCGE, Marketplace, Bonsai, Billing, Privacy — and lets the other five wait for evidence of demand. The compression preserves the manifesto, not the surface area." — SPARTAN SI × ADA · FORGE Institute · ATANDA Studio*

2. SPARTAN Compression Report — SCM 7-Step Execution Log

[Step 1] SCAN — Source Manifest

```
Source PDD:       JNGL-ARK-PDD-V3-2026
Source surfaces:  12 user-facing feature areas
Source routes:    89 HTTP endpoints
Source tables:    34 Drizzle tables (Postgres)
Source prompts:   88 (one per atomic build unit in v3 worksheet)

Phase distribution in source:
  Phase A · Foundation/auth/sessions:          12
  Phase B · Resume analyzer + JST scoring:     11
  Phase C · CCGE + KCSE + scenarios:           10
  Phase D · SPHINX core + HIVE precheck:       11
  Phase E · Roundtable + synergy + synthesis:   9
  Phase F · Forge Lab + Bonsai DAG:             8
  Phase G · Cohorts + assignments + CSV:        9
  Phase H · GUIN+ + endorsements:               5
  Phase I · Billing + Stripe stub:              7
  Phase J · ARK identity flywheel + SSE:        6

Priority:         P0=32 · P1=29 · P2=18 · P3=9
Stack:            React + Vite + Tailwind + Recharts (FE) ·
                  Express + tsx + Drizzle ORM + Postgres (BE) ·
                  Anthropic Claude (Haiku + Sonnet) · Replit autoscale
IDE / Conductor:  Replit Agent (single-developer flow)

Feature Registry (all user-facing — all classified below):
  [F-01] Identity (ARK / JST / CCMI / LHCS) + SSE stream
  [F-02] Resume Analyzer (PDF/text · 12-vector + vulnerability + pivot)
  [F-03] CCGE Arena (single-player card game · KCSE judging)
  [F-04] SPHINX × Matrix Marketplace (publish · browse · purchase)
  [F-05] Matrix Forge Lab (.docx → HIVE precheck → publish handoff)
  [F-06] Bonsai Seller Onboarding (18-stage DAG · dependency-gated)
  [F-07] Cohorts & Institutional (instructor · assignments · CSV)
  [F-08] GUIN+ Identity (public profile · Knight rank · endorsements)
  [F-09] Billing & Subscription (FREE/PRO/SCHOOL/ENTERPRISE · stub)
  [F-10] AI Surfaces (Claude · narrative · scenario gen · KCSE judge)
```

```
  [F-11] GDPR/CCPA (export + cascading delete)
  [F-12] Admin & Seed (seed, backfill, webhook simulate, DRM)
```

[Step 2] PROFILE — Classification Results

```
TOTAL CLASSIFIED: 88 prompts (12 features)

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CLASS A — PRESERVE (28 prompts · 31.8%)
  7 features that close the ARK identity loop end-to-end:
    F-01 Identity (ARK/JST/CCMI/LHCS) + SSE stream
    F-02 Resume Analyzer (with 12-vector + vulnerability + pivot)
    F-03 CCGE Arena (deterministic local judging only — Claude deferred)
    F-04 SPHINX × Matrix Marketplace (publish · browse · purchase)
    F-06 Bonsai Seller Onboarding (18-stage DAG · static stages)
    F-09 Billing & Subscription (FREE/PRO only · synthetic Stripe)
    F-11 GDPR/CCPA (export + cascade delete · compliance-required)
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CLASS B — SYNTHESISE (24 prompts · 27.3%)
  F-05 Forge Lab .docx upload  →  merged into F-04 publish flow
                                  (paste-text path acceptable for MVP)
  F-08 GUIN+ public profile     →  merged into /profile view (own page
only)
  F-10 KCSE Claude judge        →  merged into F-03 (deterministic only)
  F-12 Seed endpoint            →  merged into F-01 (dev-only bootstrap)
  ARK history endpoint          →  merged into F-01 identity payload
  ARK flywheel-cta              →  merged into dashboard render
  LHCS detail endpoint          →  merged into F-01 identity payload
  Recharts radar (12-vector)    →  merged into F-02 result page
  Vulnerability heatmap (V005)  →  merged into F-02 result page
  Vulnerability timeline (V006) →  merged into F-02 result page
  Archetype handicap viz        →  merged into F-02 result page
  Skill gap matrix (V008)       →  merged into pivot card on F-02
  SPHINX hive-precheck          →  merged into F-04 publish form
  SPHINX listings by-creator    →  merged into /profile creator block
  SPHINX credits balance        →  merged into F-04 listing card
  Endorsements POST/GET         →  CLASS C (defer; no value pre-revenue)
  AI status endpoint            →  merged into admin diagnostic page
  Notifications list/read       →  CLASS C (in-product nudge only)
  DRM violators / event         →  CLASS C (no enforcement loop pre-
revenue)
  Admin CCGE compendium import  →  CLASS C (seed covers MVP card set)
  Admin billing webhook-simulate→  merged into seed (dev-only)
  Custom CCGE scenario create   →  CLASS C (curated set ships in seed)
  Roundtable recompute / pairs  →  CLASS C (no fanout pre-100 listings)
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  Synthesis sessions (full)     →  CLASS C (multi-creator pre-revenue)
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CLASS C — DEFER (36 prompts · 40.9%)
  All with documented upgrade triggers in PART 4:
    F-07 Cohorts (instructor / students / CSV)   [trigger: school SKU]
    F-08 GUIN+ public profile pages (/u/:slug)   [trigger: 100 sellers]
    F-10 Claude narrative (Sonnet · Pro feature) [trigger: PRO billing
live]
    F-10 Custom scenario gen (admin)             [trigger: card set
exhausted]
    SPHINX Roundtable awareness graph            [trigger: 100+
listings]
    SPHINX complementary pairs API               [trigger: 100+
listings]
    SPHINX synthesis sessions (multi-creator)    [trigger: 25+ creators]
    SPHINX synergies calculate                   [trigger: 100+
listings]
    AI usage ledger UI                           [trigger: $100/mo
Claude spend]
    AI cache admin UI                            [trigger: cache hit
rate < 60%]
    Endorsements (give + display)                [trigger: 50+ sellers]
    Notifications stream + read state            [trigger: 500 DAU]
    DRM event ingest + violators                 [trigger: first
redistribution claim]
    Knight ranks (GUIN+ contributor gamif)       [trigger: 250+ active
sellers]
    /context-craft levels reference page         [trigger: post-launch
SEO]
    SCHOOL_STUDENT + ENTERPRISE plans            [trigger: cohort SKU
live]
    Stripe live webhooks (real, not simulated)   [trigger: $1K MRR]
    Subscription cancel + dunning                [trigger: first
cancellation]
    Billing receipt emails                       [trigger: PRO live]
    Bonsai DAG react-flow visualisation          [trigger: 50+ in-
progress sellers]
    Bonsai dependency badge on each stage        [trigger: support
tickets > 10]
    Card synergies cache table                   [trigger: 100+
listings]
    Complementary pairs precompute job           [trigger: Roundtable
live]
    Roundtable state snapshot table              [trigger: Roundtable
live]
    Test results dashboard                       [trigger: ≥3 QA
engineers]
    Department reference / enterprise heatmap    [trigger: first
```

```
enterprise]
    /enterprise workforce intelligence page      [trigger: first
enterprise]
    /report PDF export (jsPDF + html2canvas)     [trigger: PRO live]
    /demo public investor page (Sarah Chen)      [trigger: first
investor meeting]
    Sidebar GUIN+ identity card                  [trigger: GUIN+ profile
live]
    Resume narrative button on dashboard         [trigger: Claude
narrative live]
    Admin backfill UI (currently CLI-only)       [trigger: 1000+ users]
    Mobile-responsive Recharts variants          [trigger: mobile share
> 20%]
    SAML/OIDC SSO                                [trigger: first
enterprise RFP]
    i18n shell                                   [trigger: first non-EN
seller]
    A11y formal audit (WCAG 2.1 AA)              [trigger: pre-Series A]
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RATIOS:
  CLASS A: 31.8%  □  (target 25–40%)
  CLASS B: 27.3%  □  (all 24 merged into CLASS A rewrites — 0 extra
prompts)
  CLASS C: 40.9%  □  (target 35–60%)
```

[Step 3] ASSESS — VIBE DJ Confirmation

```
VIBE DJ RE-EVALUATION for ARK PLATFORM MVP:

The source codebase is a single TypeScript monorepo (React + Vite +
Express
+ Drizzle + Postgres) already operational on Replit. SPARTAN ASSESS
confirms
Replit Agent as the MVP conductor:

  Replit Agent:  92/100  □  PRIMARY — single-developer, single-
environment
                              flow with workflows, secrets, autoscale
deploy,
                              integration blueprints (Anthropic), and
                              checkpoint/rollback all native. Cannot be
                              substituted without losing the build-
deploy
                              parity that gets a single dev to launch.
```

```
  Cursor:        78/100      ALTERNATE — IDE-only; loses Replit's deploy
                              and secrets integration. Acceptable for
power
                              users who want local-only iteration.
  Lovable:       28/100      INCORRECT TOOL — Lovable builds new
projects
                              from scratch with its own stack opinions;
                              cannot extend the existing Express +
Drizzle
                              monorepo without a rewrite.
  v0.dev:        84/100  □  SESSION MUSICIAN — Recharts variants for
                              JST gauge, ARK identity card, Bonsai DAG
                              node, marketplace listing card. Generated
                              once before each visual sprint, exported
as
                              JSX, imported into Replit. Not the
conductor.

PRIMARY CONDUCTOR:  REPLIT AGENT (92/100) — CONFIRMED
SESSION MUSICIAN:   v0.dev (84/100) — for 4 visual components only
AVS check:          All CLASS A surfaces confirmed implementable in
Replit
                    extending the existing codebase. AVS = 100%.
```

[Steps 4–5] REDUCE + TRANSFORM

```
CLASS C removed:     36 prompts  → Upgrade Path Document (Part 4)
CLASS B synthesised: 24 prompts  → ALL merged into 24 MVP prompts (0
extra)
CLASS A rewritten:   28 prompts  → 24 MVP prompts (4 merges applied)

Key merges:
  ark.identity + ark.history + ark.flywheel-cta + ark.lhcs
                                  → MVCC-IDENT-002 (single identity
payload)
  resume.upload + 4 viz components + archetype handicap
                                  → MVCC-RES-002 (single result page)
  ccge.cards + ccge.scenarios + ccge.sessions create/finish
                                  → MVCC-CCGE-002 (single arena page +
API)
  sphinx.hive-precheck + sphinx.listings publish
                                  → MVCC-MKT-002 (publish-with-precheck
form)
  sphinx.listings GET + by-creator + credits/:userId
                                  → MVCC-MKT-001 (browse + listing
```

```
detail)
  sphinx.purchase + credit ledger
                                  → MVCC-MKT-003 (transactional
purchase)
  bonsai.progress GET + complete + 18 stages constant
                                  → MVCC-ONB-001 (single onboarding
page)
  billing.me + checkout + complete + cancel
                                  → MVCC-BILL-001 (single billing flow)
  users.me/export + DELETE users/me
                                  → MVCC-PRIV-001 (single privacy page)
  seed + admin webhook-simulate
                                  → MVCC-DEV-001 (single dev bootstrap
route)

Net MVP prompts:  24
CR_p = (88 − 24) / 88 = 72.7%   □  CERTIFIED  (50–80% range)
CR_t = (12 − 4) / 12 = 66.7%    □  (12 weeks source → 4 weeks MVP)
CR_c   infra unchanged (~$0 delta — same Replit autoscale · same
Postgres
       · Claude usage capped via ai_usage budget enforcement)
```

[Step 6] ZPOS+5 Application

CONTENT TYPE	METHOD	AVG REDUCTION	SEMANTIC PRES.
P0 core prompts (identity flywheel · billing · privacy)	PRISM	38–42%	96.8%
Route + SSE technical (Express handlers + EventSource)	QUANTUM	40–46%	95.4%
Worksheet rows + classification tables	SYNTHESIS	35–39%	96.7%
OVERALL AVERAGE	Mixed	**40.1%**	**96.3%**

[Step 7] PACKAGE — Quality Gate Results

GATE	SCORE	STATUS
FFS — Feature Fidelity	100%	□ All 7 CLASS A features present end-to-end
AVS — Architecture Viability	100%	□ All MVP prompts implementable in Replit Agent
UIS — Upgrade Integrity	100%	□ All 36 CLASS C deferrals carry triggers

CR_p — Compression Ratio	**72.7%**	▯ **CERTIFIED** (50–80% range)
SPS — Semantic Preservation	96.3%	▯ (≥95% required)

```
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            SPARTAN COMPRESSION CERTIFICATE
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Cert ID:        JNGL-ARK-MVP-2026-001
Status:         CERTIFIED
FFS:            100%   ▯
AVS:            100%   ▯
UIS:            100%   ▯
CR_p:           72.7%  ▯  CERTIFIED  (target 50–80%)
CR_t:           66.7%  ▯  (12 wk → 4 wk)
SPS:            96.3%  ▯  (≥ 95% required)
CLASS A:        28 prompts (→ 24 MVP prompts via 4 merges)
CLASS B:        24 prompts (all merged into CLASS A rewrites)
CLASS C:        36 prompts (deferred with documented triggers)
ZPOS+5:         PRISM · QUANTUM · SYNTHESIS
Avg reduction:  40.1%  |  Semantic: 96.3%
Target VIBE:    Replit Agent (92/100)
Session Music:  v0.dev (84/100)
Compressor:     SPARTAN SI v1.0.0 (SPRT-ATLAS-MVP-SI-2026-001)
Signed:         May 2026
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```

3. Part 1 — VIBE DJ Analysis

PRIMARY CONDUCTOR: REPLIT AGENT (92/100 · HIGH · TARANTULA mandate)

The ARK Platform codebase already lives on Replit — a single-developer, single-environment monorepo with workflows, autoscale deploy, Anthropic integration, and checkpoint/rollback all native. The conductor cannot be substituted without losing that build-deploy parity. Cursor is acceptable as an alternate IDE for power users but loses the deploy and secrets surface. Lovable is the wrong tool entirely (it cannot extend an existing Express + Drizzle monorepo). The 24 MVP prompts are written for Replit Agent and reference its workflows, secrets, and integrations vocabulary throughout.

SESSION MUSICIAN: v0.dev (84/100) — generates four visual components once before the visual sprints, exported as JSX, imported into Replit: (1) JST gauge with five colour zones (VIZ-001), (2) ARK identity card with ARK + JST + CCMI sub-bars (Phase J), (3) Bonsai DAG node (4 visual states), (4) marketplace listing card with HIVE chip + price + synergy hints. All four use the project's HSL palette (cyan / emerald / crimson) and Orbitron / Rajdhani / Space Grotesk font stack.

Why not Cursor as primary? ARK's deploy story is Replit autoscale and ARK's secrets story is the Replit integration blueprint for Anthropic. Cursor as primary would require parallel deploy and secrets surfaces — adding two pieces of infrastructure for zero MVP benefit.

Why no other tools? No SaaS layer is needed. Stripe is stubbed for MVP (real Stripe is a CLASS C trigger). Anthropic is already integrated via Replit blueprint. Postgres is already auto-provisioned via `DATABASE_URL`. The MVP introduces no new SaaS dependencies.

4. Part 2 — Architecture Reduction (Stack Collapse Map)

4.1 What STAYS — 7 Features (FFS = 100%)

FEATURE	PRODUCTION IMPLEMENTATION (PDD V3)	MVP (REPLIT AGENT · MERGED)	STATUS
Identity (ARK/JST/CCMI/LHCS)	`arkRecalc` + 6 endpoints + SSE	MVCC-IDENT-001/002/003 (recalc + single payload + SSE)	□ Full
Resume Analyzer	`resumeAnalyzer` + 4 viz components + archetype	MVCC-RES-001/002 (analyzer + single result page)	□ Full
CCGE Arena	7 endpoints + KCSE Claude judge + scenarios	MVCC-CCGE-001/002 (deterministic only · curated scenarios)	□ Full (Claude deferred)
SPHINX × Matrix Marketplace	18 endpoints + Roundtable + synthesis	MVCC-MKT-001/002/003 (browse · publish · purchase only)	□ MVP (graph deferred)
Bonsai Seller Onboarding	18-stage DAG + react-flow viz	MVCC-ONB-001 (single page · list view · dependency-gated)	□ Full (DAG viz deferred)
Billing & Subscription	6 endpoints + 4 plans + Stripe stub	MVCC-BILL-001 (FREE + PRO only · synthetic)	□ MVP
GDPR/CCPA	export + cascade delete	MVCC-PRIV-001 (single page · both flows)	□ Full

4.2 CLASS B — All 24 Merged (0 Extra Prompts)

PRODUCTION PROMPT	SYNTHESIS	MERGED INTO
ARK history endpoint	Returned in identity payload (last 30 days inline)	MVCC-IDENT-002

ARK flywheel-cta endpoint	Computed at render from identity payload	MVCC-IDENT-002
ARK lhcs detail endpoint	LHCS detail already in identity payload	MVCC-IDENT-002
Recharts 12-vector radar	One Recharts radar component on result page	MVCC-RES-002
Vulnerability heatmap (V005)	Recharts heatmap component on result page	MVCC-RES-002
Vulnerability timeline (V006)	Recharts AreaChart on result page	MVCC-RES-002
Archetype handicap viz	Animated bars + trait tags inline	MVCC-RES-002
Skill gap matrix (V008)	Table inside each pivot card	MVCC-RES-002
KCSE Claude judge	Deterministic local scoring only (Claude is CLASS C)	MVCC-CCGE-002
Forge Lab .docx upload	Paste-text path replaces upload for MVP	MVCC-MKT-002
HIVE precheck endpoint	Inline preview button on publish form	MVCC-MKT-002
SPHINX listings by-creator	Block on /profile (own listings only)	MVCC-PROF-001
SPHINX credits balance	Read inline in listing card and purchase modal	MVCC-MKT-001
AI status endpoint	Read inline in dev/admin diagnostic route	MVCC-DEV-001
Admin webhook-simulate	Dev-only convenience inside seed route	MVCC-DEV-001
Seed endpoint (full v3 set)	Trimmed seed: 10 cards + 6 scenarios + 1 demo user	MVCC-DEV-001
GUIN+ public profile (/u/:slug)	Own profile only; public discovery → CLASS C	MVCC-PROF-001
Endorsements give/display	→ CLASS C (no value pre-revenue)	—
Roundtable / pairs / synergies	→ CLASS C (requires 100+ listings)	—
Synthesis sessions	→ CLASS C (multi-creator pre-revenue)	—
Notifications	→ CLASS C (in-product nudge only)	—
DRM event / violators	→ CLASS C (no enforcement loop pre-revenue)	—
Custom CCGE scenario create	→ CLASS C (curated set covers MVP)	—
Admin CCGE compendium import	→ CLASS C (seed covers MVP card set)	—

4.3 CLASS C — 36 Deferrals (All Upgrade-Triggered)

DEFERRED SURFACE	TRIGGER	EFFORT
Cohorts (instructor / students / CSV)	First SCHOOL_STUDENT licence sold	4 days
GUIN+ public profile pages (/u/:slug)	100+ sellers OR first cited profile	2 days
Claude narrative on dashboard	PRO billing live + $100/mo Claude budget cleared	1 day
Custom scenario generation (admin)	Curated set engagement < 30%	2 days
SPHINX Roundtable awareness graph	100+ listings	3 days
SPHINX complementary pairs API	Roundtable live	2 days
SPHINX synthesis sessions	25+ creators OR first co-authoring request	5 days
AI usage ledger UI	Claude spend > $100/mo	1 day
AI cache admin UI	Cache hit rate < 60% (currently > 80%)	1 day
Endorsements (give + display)	50+ active sellers	2 days
Notifications stream + read state	500+ daily active users	3 days
DRM event ingest + violators	First redistribution claim	2 days
Knight ranks (GUIN+ gamification)	250+ active sellers	2 days
Context-craft levels reference page	Post-launch SEO push	1 day
SCHOOL_STUDENT + ENTERPRISE plans	Cohort SKU live	3 days
Stripe live webhooks (real, not stub)	$1K MRR	4 days
Subscription cancel + dunning	First cancellation logged	2 days
Billing receipt emails (Resend)	PRO live	1 day
Bonsai DAG react-flow viz	50+ in-progress sellers OR support tickets > 10	3 days
Bonsai dependency badge per stage	Support tickets on "what's blocking me?" > 5	1 day
Card synergies cache table	100+ listings	1 day
Complementary pairs precompute job	Roundtable live	1 day
Roundtable state snapshot table	Roundtable live	1 day
Test results dashboard	≥3 QA engineers	2 days
Department reference + enterprise heatmap	First enterprise subscriber	3 days
/enterprise workforce intelligence page	First enterprise subscriber	4 days

/report PDF export (jsPDF + html2canvas)	PRO live + first PDF request	2 days
/demo public investor page (Sarah Chen)	First investor meeting scheduled	2 days
Sidebar GUIN+ identity card	GUIN+ public profile live	1 day
Admin backfill UI (currently CLI-only)	1000+ users in database	1 day
Mobile-responsive Recharts variants	Mobile share > 20%	3 days
SAML/OIDC SSO	First enterprise RFP requirement	5 days
i18n shell	First non-EN seller request	4 days
A11y formal audit (WCAG 2.1 AA)	Pre-Series A diligence	5 days
Resume narrative button	Claude narrative live	0.5 day
GUIN+ contributor leaderboards	Knight ranks live	1 day

4.4 MVP at a Glance — 24 Prompts

```
PHASE 1 — FOUNDATION & IDENTITY        (5 prompts · Week 1)
  MVCC-DB-001     Drizzle schema (22 tables) + idempotent migration +
indexes
  MVCC-AUTH-001   Express session + bcrypt + helmet + rate limits +
requireAuth/Self
  MVCC-IDENT-001  scoringEngine.ts (JST + CCMI + ARK math, pure, unit-
tested)
  MVCC-IDENT-002  arkRecalc.ts (single writer · caps · LHCS composite ·
history)
  MVCC-IDENT-003  Orchestrator event bus + SSE stream + useArkStream
client hook

PHASE 2 — RESUME LOOP                  (4 prompts · Week 2)
  MVCC-RES-001    resumeAnalyzer (6 categories · 14 vuln patterns ·
archetype map)
  MVCC-RES-002    /upload (5-phase UI) + /dashboard (identity + JST +
radar + vuln
                  + heatmap + timeline + archetype + 12-vector + pivot +
upskill)
  MVCC-DEV-001    POST /api/seed (dev-only · 10 cards · 6 scenarios · 1
demo user
                  · webhook-simulate · ai-status diagnostic — all
merged)
  MVCC-DESIGN-001 v0.dev pre-gen: JST gauge + ARK card + Bonsai node +
listing card
```

```
PHASE 3 — CCGE + FLYWHEEL              (4 prompts · Week 3 early)
  MVCC-CCGE-001   ccge.ts engine (curated 14 cards + 6 scenarios ·
deterministic
                  KCSE scoring · transactional finalize)
  MVCC-CCGE-002   /play arena page (deal 5 · write hand · finalize ·
KCSE breakdown)
  MVCC-FLY-001    Flywheel caps enforcement (+15/d CCGE · +20/30d SPHINX
· clamp
                  rounding overshoot off CCMI · downward-sync for admin
recalc)
  MVCC-ONB-001    /onboarding page (18-stage list view · dependency-
gated complete
                  · zero ARK/HIVE side-effects · server-side dep
validation)

PHASE 4 — MARKETPLACE MVP              (5 prompts · Week 3 late + Week 4
early)
  MVCC-MKT-001    SPHINX browse + listing detail (paginated ·
pillar/status filter
                  · credits read inline · purchase modal trigger)
  MVCC-MKT-002    SPHINX publish form (CC_400 gate · HIVE precheck
inline preview
                  · paste-text body · price/title/pillar/description)
  MVCC-MKT-003    Transactional purchase (debit credits · transfer
access · award
                  ARK delta · ledger entry · row-level locking)
  MVCC-PROF-001   /profile page (editable details · own listings block ·
purchases
                  · sales · credits balance · GUIN+ identity card — own
only)
  MVCC-MKT-004    HIVE precheck engine (14-dim scoring ·
Bronze/Silver/Gold/Plat
                  · publish gate 80 = Gold = CC_400 floor)

PHASE 5 — BILLING · PRIVACY · LAUNCH    (6 prompts · Week 4 late)
  MVCC-BILL-001   Billing flow (FREE + PRO only · synthetic Stripe ·
checkoutSessions
                  · billing_events audit log · /subscription +
/checkout/:id pages)
  MVCC-PRIV-001   /privacy + /terms + /profile data-privacy section (GET
export ·
                  DELETE me with confirm:"DELETE" · single transactional
cascade)
  MVCC-QA-001     Smoke test suite (resume upload → identity recalc →
CCGE finish
                  → flywheel cap → publish → purchase → export → delete)
  MVCC-SEC-001    Security audit (8 vectors: session regen, role-strip
on PUT,
```

```
                  CC cert 403, requireSelf, body limits, rate limits,
HIVE gate,
                  transactional purchase) + helmet CSP for production
  MVCC-LAUNCH-001 Production checklist (SESSION_SECRET required · seed
403 in prod
                  · Anthropic blueprint health · autoscale deploy · 1KB
ai budget)
  MVCC-DOC-001    replit.md + threat_model.md + CHANGELOG.md (Phase J
entry only)
```

5. Part 3 — Compressed MVP Worksheet

> ***Cadence convention:*** *every MVP prompt below is single-developer-completable in 1–3 hours with Replit Agent. Camelot lineage names the two FORGE archetypes that lead each prompt. ZPOS shows the dominant compression method applied. JCSE is the projected ATLAS-rubric quality score after one revision pass.*

5.1 ▯ PHASE 1 — FOUNDATION & IDENTITY · Week 1

MVCC-DB-001 · Drizzle schema (22 tables) + idempotent migration

Camelot: ADA × TARANTULA | **ZPOS:** PRISM 41% | **JCSE:** 50/50 | **Deps:** —

Single `shared/schema.ts` declaring 22 Drizzle tables: `users`, `assessments`, `upskilling_plans`, `pivot_opportunities`, `transferability_vectors`, `jnomics_cards`, `ccge_cards`, `ccge_scenarios`, `game_sessions`, `spc_listings`, `spc_purchases`, `user_credits`, `checkout_sessions`, `billing_events`, `ark_events`, `ai_usage`, `ai_cache`, `ccmi_pillar_scores`, `ark_score_history`, `lhcs_signals`, `platform_credit_ledger`, `bonsai_progress`. Plus `SCORE_GLOSSARY`, `JST_WEIGHTS`, `CCMI_WEIGHTS`, `KCSE_WEIGHTS`, `JCSE_TIER_THRESHOLDS`, `HIVE_TIER_THRESHOLDS`, `LHCS_WEIGHTS`, `LHCS_BANDS`, `FLYWHEEL_CAPS`, `CONTEXT_CRAFT_HANDICAP`, `SPC_MIN_CERT_TO_PUBLISH`, `JST_ARCHETYPE_MAP`. Migration `0000_mvp_init.sql` is `CREATE TABLE IF NOT EXISTS` + `ADD COLUMN IF NOT EXISTS` throughout. Indexes on `users(email)`, `assessments(user_id, created_at desc)`, `game_sessions(user_id)`, `spc_listings(creator_id, status)`, `spc_purchases(buyer_id)`, `ark_events(user_id, created_at)`.

MVCC-AUTH-001 · Express + session + bcrypt + helmet + rate limits

Camelot: TARANTULA × ADA | **ZPOS:** PRISM 39% | **JCSE:** 49/50 | **Deps:** MVCC-DB-001 `server/index.ts`: Express on port 5000, helmet with prod CSP / HSTS / `X-Frame-Options: SAMEORIGIN` / COOP / CORP / `Referrer-Policy: no-referrer`, body limit 1 MB. `connect-pg-simple` session store, cookie `ark.sid` (httpOnly, sameSite=lax, 14-day rolling), refuses to boot if `SESSION_SECRET` unset in production. `server/auth.ts`: `requireAuth`, `requireSelf(:param)`, `loginSession(req, user)` regenerates session id. Bcrypt cost 10 at storage boundary with transparent legacy-plaintext rehash on first successful login. Rate limits: 240 req/min global, 20 failed auths / 15 min per IP. `POST /api/auth/register` (auto-login), `POST /api/auth/login` (regen session), `POST /api/auth/logout`, `GET /api/auth/me`.

MVCC-IDENT-001 · scoringEngine.ts (pure math)

Camelot: ADA × ATLAS | **ZPOS:** PRISM 42% | **JCSE:** 50/50 | **Deps:** MVCC-DB-001

Pure, side-effect-free `server/scoringEngine.ts`. `computeJst({jobs,skills,talent,certLevel}) → {jst, jstRaw, modifier}` applying `(J·.30+S·.40+T·.30)·3` and `CONTEXT_CRAFT_HANDICAP[certLevel]`. `computeCcmi(pillarScores) → ccmi` applying weighted P1–P7 sum · 3. `computeArk(jst, ccmi) → {ark, capped, jst, ccmi}` enforcing 0–600 cap, hard-clamping rounding overshoot off **CCMI** side. `arkId(userId) → string` (SHA-256 first-12-hex of `userId + epoch`). Unit tests under `tests/scoring.test.ts` covering: weight invariants, cap invariants, modifier ladder, overshoot direction, ARK-ID stability.

MVCC-IDENT-002 · arkRecalc.ts (single writer)

Camelot: ADA × TARANTULA | **ZPOS:** PRISM 41% | **JCSE:** 50/50 | **Deps:** MVCC-IDENT-001

`server/arkRecalc.ts::recalcArkForUser(userId, trigger, delta?)` is the **only** writer of identity fields. Inside one DB transaction: read current `users` + `ccmi_pillar_scores` + `lhcs_signals`, compute new JST + CCMI + ARK via scoringEngine, apply `FLYWHEEL_CAPS` (CCGE +15/day, SPHINX +20/30d), hard-clamp overshoot off CCMI, write `users` + `ccmi_pillar_scores` + `lhcs_signals` + append `ark_score_history` row. Trigger types: `assessment.completed`, `ccge.session.finalized`, `sphinx.purchase`, `manual.recompute` (never positive), `backfill` (never positive). LHCS composite: `round(0.35·CPR + 0.35·MPS + 0.30·LCIS)`; status is the threshold band of the composite (`≥70 ACTIVE / ≥40 DEVELOPING / <40 BASELINE`), not a roll-up of the three lights.

Endpoints `GET /api/ark/identity` (returns identity + last-30-day history + flywheel-cta + lhcs detail in one payload), `POST /api/ark/recalc` .

MVCC-IDENT-003 · Orchestrator + SSE + useArkStream

Camelot: EVE × TARANTULA | **ZPOS:** QUANTUM 44% | **JCSE:** 49/50 | **Deps:** MVCC-IDENT-002

`server/orchestrator.ts` : typed `EventEmitter` keyed by event name (`assessment.completed` , `ccge.session.finalized` , `sphinx.purchase` , `ark.identity`). Each emit triggers `recalcArkForUser` then publishes `ark.identity` to per-user SSE channels. `GET /api/ark-score/stream` opens an SSE per session, sends initial snapshot, then live merges. `GET /api/ark-score/events` is a poll fallback (last 20 events). Client hook `client/src/lib/useArkStream.ts` : opens EventSource with `credentials: include` , exponential-backoff reconnect (1s → 30s cap), snapshot-merge on each tick, exposes `{identity, isLive, lastEventAt}` .

5.2 PHASE 2 — RESUME LOOP · Week 2

MVCC-RES-001 · resumeAnalyzer (6 categories · vulnerability · archetype)

Camelot: ATLAS × ADA | **ZPOS:** PRISM 43% | **JCSE:** 50/50 | **Deps:** MVCC-IDENT-001

`server/resumeAnalyzer.ts` : accepts PDF (via `pdf-parse` v2 `new PDFParse({data}).getText()` , 15s bounded race) or plain text (10 MB cap). Six weighted keyword categories (technical, leadership, analytical, communication, innovation, AI-adjacent) score JST sub-dimensions. 14 regex automation-risk patterns produce a 0–4 vulnerability level adjusted by AI/leadership scores. Archetype handicap: three signals — title-from- `JST_ARCHETYPE_MAP` (228+ titles · 40%), skill-category weights (35%), matched FORGE card classifications (25%) — normalised, top archetype sets `readinessProfile` (Architect / Orchestrator / Conductor). Generates exactly 12 transferability vectors, 3 pivot opportunities, 3 upskilling plans. Endpoints: `POST /api/resume/upload` (multipart), `POST /api/assessments` (manual), `POST /api/assessment/text` (text-only), `GET /api/assessments/user/:userId/latest` .

MVCC-RES-002 · /upload + /dashboard pages (single result surface)

Camelot: TARANTULA × v0.dev | **ZPOS:** SYNTHESIS 37% | **JCSE:** 48/50 | **Deps:** MVCC-RES-001, MVCC-IDENT-003, MVCC-DESIGN-001

`client/src/pages/upload.tsx` : dropzone (PDF or paste-text), 5-phase progress UI (Discovery & Extraction → JST Calc → Vulnerability → Transferability → Recommendations), routes to `/dashboard` on success. `client/src/pages/dashboard.tsx` : ARK identity card (from v0.dev) at top, then 8 component blocks: JST gauge (VIZ-001, five colour zones, percentile, industry avg, trend), JST radar (VIZ-002, three-layer with industry overlay), vulnerability meter (VIZ-004, five levels), task heatmap (VIZ-005, time allocation), vulnerability timeline (VIZ-006, AreaChart to 2041), archetype handicap (animated bars + trait tags), 12-vector transferability radar (VIZ-007), 3 pivot opportunity cards with embedded skill-gap matrix (VIZ-008). All Recharts. All live-merged via `useArkStream` .

MVCC-DEV-001 · Dev bootstrap (seed + webhook-simulate + ai-status)

Camelot: TARANTULA × ADA | **ZPOS:** SYNTHESIS 38% | **JCSE:** 47/50 | **Deps:** MVCC-AUTH-001, MVCC-DB-001

`POST /api/seed` (returns 403 in production via `NODE_ENV === "production"` check): inserts 10 FORGE cards (`card-001` – `card-010`), 14 CCGE cards, 6 CCGE scenarios (2 each Bronze/Silver/Gold), 1 demo user `analyst@enterprise.com` / `arkplatform` with a fully-populated assessment, 1 demo creator `creator@sphinx.io` (CC_500) with 3 sample SPCs. Sub-route `POST /api/seed/webhook-simulate` simulates a Stripe `checkout.session.completed` . `GET /api/ai/status` returns Anthropic blueprint health, current `ai_usage` daily total, and cache hit rate.

MVCC-DESIGN-001 · v0.dev component pre-generation (4 components)

Camelot: v0.dev × TARANTULA | **ZPOS:** PRISM 40% | **JCSE:** 48/50 | **Deps:** —

Generated once on v0.dev, exported as JSX, dropped into `client/src/components/` under explicit `// v0.dev generated` headers: (1) `ArkIdentityCard.tsx` — top-of-dashboard, shows ARK + JST + CCMI sub-bars, ARK-ID string, SSE live chip; (2) `JstGauge.tsx` — semicircular five-zone gauge with Orbitron numeric; (3) `BonsaiStageNode.tsx` — 4 visual states (completed / current / locked / available) with phase badge; (4) `MarketplaceListingCard.tsx` — title, HIVE chip, price, synergy hint chip, purchase CTA. All four use the project palette (`hsl(188 86% 53%)` cyan / `hsl(152 69% 31%)` emerald / `hsl(346 87% 43%)` crimson) and the Orbitron / Rajdhani / Space Grotesk font stack.

5.3 ▯ PHASE 3 — CCGE + FLYWHEEL · Week 3 early

MVCC-CCGE-001 · ccge.ts engine (deterministic KCSE scoring)

Camelot: ADA × ATLAS | **ZPOS:** PRISM 40% | **JCSE:** 49/50 | **Deps:** MVCC-DB-001, MVCC-IDENT-002

`server/ccge.ts` : card catalog (14 cards seeded), scenario catalog (6 scenarios tiered Bronze/Silver/Gold), session lifecycle. `POST /api/ccge/sessions` deals 5 cards from a tier-appropriate deck. `POST /api/ccge/sessions/:id/finish` takes `{handCardIds, response}` , runs **deterministic** KCSE scoring (no Claude in MVP): Knowledge from required-card-coverage, Clarity from response length + structural markers, Specificity from concrete-noun density, Efficiency from token-economy ratio. Composite JCSE = `K·.30 + C·.30 + S·.20 + E·.20` . Transactional finalize: insert `game_sessions` row, emit `ccge.session.finalized` (orchestrator triggers `recalcArkForUser` with CCGE cap +15/day).

MVCC-CCGE-002 · /play arena page

Camelot: TARANTULA × v0.dev | **ZPOS:** SYNTHESIS 38% | **JCSE:** 48/50 | **Deps:** MVCC-CCGE-001

`client/src/pages/play.tsx` : tier selector (Bronze / Silver / Gold), scenario card with prompt brief and KCSE rubric reminder, 5-card hand with drag-to-reorder, response textarea, "Submit Hand" button. On submit shows KCSE breakdown (four bars, Knowledge / Clarity / Specificity / Efficiency), JCSE composite, tier badge (Bronze 30 / Silver 36 / Gold 43 / Platinum 48), ARK delta chip ("+12 ARK" with cap-status footnote). Live identity card at top updates via SSE.

MVCC-FLY-001 · Flywheel caps enforcement

Camelot: ADA × TARANTULA | **ZPOS:** PRISM 41% | **JCSE:** 50/50 | **Deps:** MVCC-IDENT-002

`server/arkRecalc.ts::applyCaps(userId, trigger, intendedDelta) → {finalDelta, intendedCap, reason}` : looks up `ark_events` in the trigger's window (1 day for CCGE, 30 days for SPHINX), sums prior positive deltas, returns the remaining headroom. Recalc enforces `finalDelta ≤ intendedCap` . Triggers `manual.recompute` and `backfill` can never award positive ARK (downward-sync only — clamps `finalDelta = min(finalDelta, 0)`). Rounding overshoot from proportional JST/CCMI scaling is hard-clamped off the **CCMI** side before persistence (preserves the "career capital" reading of JST). Unit tests: per-day CCGE cap, per-30d SPHINX cap, downward-sync invariant, overshoot direction, idempotency of repeated recalc.

MVCC-ONB-001 · /onboarding (18-stage Bonsai · list view)

Camelot: EVE × ADA | **ZPOS:** SYNTHESIS 37% | **JCSE:** 47/50 | **Deps:** MVCC-DB-001, MVCC-AUTH-001

`shared/bonsaiStages.ts` declares 18 static stages in 4 phases (Roots 1–3, Trunk 4–7, Branches 8–11, Canopy 12–18); each stage has `{id, phase, title, goal, actions[], dependsOn[]}`. `server/bonsai.ts`: `GET /api/sphinx/bonsai/progress` returns the user's `bonsai_progress` rows; `POST /api/sphinx/bonsai/progress/:stageId/complete` verifies all `dependsOn` are completed (returns 409 otherwise), inserts a row, **emits zero ARK/HIVE side-effects** (pure pedagogy). `client/src/pages/onboarding.tsx`: list view (DAG visualisation deferred to CLASS C), four phase sections, each stage card shows title + goal + action checklist + dep status + Mark-Complete + Next-Stage controls.

5.4 ▯ PHASE 4 — MARKETPLACE MVP · Week 3 late + Week 4 early

MVCC-MKT-001 · SPHINX browse + listing detail

Camelot: ATLAS × TARANTULA | **ZPOS:** PRISM 39% | **JCSE:** 48/50 | **Deps:** MVCC-DB-001, MVCC-AUTH-001

`GET /api/sphinx/listings` (public): `?pillar=` and `?status=` filters, pagination (page/limit), returns DTO with `{id, title, pillar, hiveScore, hiveTier, price, creatorPublicName, creatorCertLevel}` — **never the body**. `GET /api/sphinx/listings/:id`: same DTO; body included only if buyer has a `spc_purchases` row or is the creator. `GET /api/sphinx/credits/:userId` (requireSelf): user's credit balance. `client/src/pages/marketplace.tsx`: grid of `MarketplaceListingCard` (from v0.dev) with filter sidebar (pillar P1–P7, HIVE tier, price range). `client/src/pages/marketplace-listing.tsx`: full detail with creator chip linking to own `/profile` (CLASS C: link to public `/u/:slug`), purchase modal trigger.

MVCC-MKT-002 · SPHINX publish form (CC_400 gate + HIVE precheck inline)

Camelot: ADA × TARANTULA | **ZPOS:** PRISM 41% | **JCSE:** 49/50 | **Deps:** MVCC-MKT-004

`POST /api/sphinx/listings` (gated on `user.contextCraftCertLevel ≥ SPC_MIN_CERT_TO_PUBLISH = CC_400`; returns 403 below): Zod-validated body `{title, pillar, price, description, body}`. Calls `runHivePrecheck(body)` server-side; if HIVE composite < 80, returns 422 with the 14-dim breakdown so the user can iterate. On pass, inserts `spc_listings` row with status `ACTIVE`. `client/src/pages/marketplace-publish.tsx`: form with title / pillar / price / description / body, "Run HIVE Precheck" button (inline preview without committing), 14-dim breakdown render, "Publish" button (disabled until precheck passes ≥ 80). The .docx upload path (Forge Lab) is CLASS C — MVP uses paste-text only.

MVCC-MKT-003 · Transactional purchase

Camelot: ADA × TARANTULA | **ZPOS:** PRISM 42% | **JCSE:** 50/50 | **Deps:** MVCC-MKT-001, MVCC-IDENT-002

`POST /api/sphinx/listings/:id/purchase` (requireAuth): single DB transaction with `SELECT ... FOR UPDATE` on `spc_listings` row and on buyer's `user_credits` row. Re-checks: listing active, buyer has not already purchased, buyer credits ≥ price, buyer is not creator. Debits credits, inserts `spc_purchases`, appends `platform_credit_ledger` (debit buyer · credit creator at platform fee net), appends `ark_events` (creator: SPHINX trigger, intendedDelta from price-tier table), emits `sphinx.purchase` (orchestrator triggers `recalcArkForUser` for creator with SPHINX cap +20/30d). Idempotency: returns 409 on duplicate purchase. All-or-nothing rollback on any failure.

MVCC-PROF-001 · /profile page (own only · GUIN+ card · marketplace blocks)

Camelot: TARANTULA × v0.dev | **ZPOS:** SYNTHESIS 36% | **JCSE:** 47/50 | **Deps:** MVCC-AUTH-001, MVCC-MKT-001

`client/src/pages/profile.tsx`: own profile only (public `/u/:slug` deferred). Sections: (1) Editable details (name, headline, link out) via `PUT /api/users/:id/profile` — **strips `role` server-side**; (2) GUIN+ identity card (Knight rank deferred to CLASS C, but CCMI breakdown + cert level shown); (3) My listings (calls `GET /api/sphinx/listings/by-creator/:userId`); (4) My purchases (`GET /api/sphinx/purchases/:userId`); (5) My sales (`GET /api/sphinx/sales/:userId`); (6) Credits balance (`GET /api/sphinx/credits/:userId`); (7) Data privacy section (links to MVCC-PRIV-001 export/delete actions). `PUT /api/users/:id/context-craft-cert` returns **403 permanently** — cert is flywheel-only.

MVCC-MKT-004 · HIVE precheck engine

Camelot: ATLAS × ADA | **ZPOS:** PRISM 43% | **JCSE:** 50/50 | **Deps:** MVCC-DB-001 `server/sphinx.ts::runHivePrecheck(body) → {dimensions: HiveDimensions, composite, tier, passed}`. Fourteen dimensions scored 0–100 each: pillar fit, constraint specificity, output schema clarity, failure mode coverage, role assignment, context richness, example presence, edge-case handling, compression ratio, semantic preservation, idempotency, observability hooks, security posture, IP cleanliness. Composite = mean(14). Tier from `HIVE_TIER_THRESHOLDS` (Bronze 60 / Silver 70 / Gold 80 / Platinum 90). `passed` = composite ≥ 80 (publish gate). Unit tests for each dimension; deterministic and bounded (under 200 ms p99).

5.5 □ PHASE 5 — BILLING · PRIVACY · LAUNCH · Week 4 late

MVCC-BILL-001 · Billing flow (FREE + PRO only · synthetic Stripe)

Camelot: POSTMIXTA × TARANTULA | **ZPOS:** PRISM 40% | **JCSE:** 48/50 | **Deps:** MVCC-AUTH-001

`server/billing.ts`: synthetic Stripe stub. `GET /api/billing/me` returns user's plan, subscription status, current period end. `POST /api/billing/checkout` creates a `checkout_sessions` row with synthetic ID, returns checkout URL `/checkout/:id`. `GET /api/billing/checkout/:id` returns the session for the form. `POST /api/billing/checkout/:id/complete` (fake card form succeeds; appends `billing_events` audit row, updates `users.plan` + `users.subscriptionStatus`). `POST /api/billing/cancel` flags the subscription cancelled at period end. Plans for MVP: **FREE** (resume upload + 1 CCGE session/day) and **PRO** ($29/mo: unlimited CCGE, narrative deferred, marketplace publish). SCHOOL_STUDENT + ENTERPRISE deferred to CLASS C.

MVCC-PRIV-001 · GDPR/CCPA (export + cascade delete)

Camelot: ADA × TARANTULA | **ZPOS:** SYNTHESIS 39% | **JCSE:** 49/50 | **Deps:** MVCC-AUTH-001

`server/storage.ts::exportUserData(userId)` returns a full JSON dump of the user's data graph (users, assessments, game_sessions, spc_listings owned, spc_purchases, ark_events, ccmi_pillar_scores, lhcs_signals, ark_score_history, bonsai_progress, checkout_sessions, billing_events). `server/storage.ts::deleteUserCascade(userId)` performs single-transaction cascade through every owned table; SPHINX listings owned by the user are anonymised rather than deleted (preserves buyer access). Endpoints: `GET /api/users/me/export`, `DELETE /api/users/me` with required body `{`

`"confirm": "DELETE" }`. Static pages `/privacy` and `/terms` describe both flows. Footer link from `AppLayout`.

MVCC-QA-001 · Smoke test suite (8 critical paths)

Camelot: TARANTULA × EVE | **ZPOS:** QUANTUM 45% | **JCSE:** 47/50 | **Deps:** all prior

Vitest smoke suite under `tests/mvp/`. Eight paths: (1) register → login → /api/auth/me; (2) resume upload (text) → /api/ark/identity reflects new JST in < 2 s; (3) CCGE finish → identity delta ≤ 15; (4) CCGE finish twice in one day → second delta clamped to remaining cap; (5) marketplace publish below HIVE 80 → 422; above → ACTIVE; (6) purchase → buyer credits debited, creator ARK delta ≤ 20 in 30 d; (7) GDPR export returns all expected keys; (8) GDPR delete cascades cleanly (no orphaned rows). All paths run under 30 s total on the Replit free tier.

MVCC-SEC-001 · Security audit (8 vectors)

Camelot: EVE × TARANTULA | **ZPOS:** PRISM 42% | **JCSE:** 50/50 | **Deps:** all prior

Inline audit checklist verified by code review pass and one Vitest case per vector: (1) Login regenerates session id (no session-fixation); (2) `PUT /api/users/:id/profile` strips `role`; (3) `PUT /api/users/:id/context-craft-cert` returns 403 permanently; (4) `requireSelf` blocks cross-user reads on assessments/credits/sales/purchases/export; (5) JSON body limit 1 MB enforced; resume 10 MB; (6) Rate limits applied (240/min global, 20 failed auths/15 min); (7) HIVE publish gate 80 enforced server-side, not client-side; (8) Purchase is atomic with `SELECT ... FOR UPDATE` and idempotent on retry. Helmet CSP locked for production: `default-src 'self'; script-src 'self'; style-src 'self' 'unsafe-inline'; img-src 'self' data:; connect-src 'self'`.

MVCC-LAUNCH-001 · Production launch checklist

Camelot: TARANTULA × POSTMIXTA | **ZPOS:** SYNTHESIS 38% | **JCSE:** 48/50 | **Deps:** all prior

20-item launch checklist run before promoting to autoscale: `SESSION_SECRET` set in production env; `DATABASE_URL` auto-provisioned; Anthropic blueprint integrated and `GET /api/ai/status` green; `NODE_ENV=production` so `/api/seed` returns 403; helmet CSP active; HSTS on; cookies `secure: true` in production; rate limits active; bcrypt cost 10 verified; 8 smoke tests green; 8 security tests green; ARK-event audit logging verified for all three flywheel triggers; checkout `success` URL whitelisted; demo user removed from production seed; replit.md current (Phase J + MVP); threat_model.md current; CHANGELOG.md MVP entry written; LICENSE present; README quickstart copy verified; first deploy tagged `mvp-v1.0.0`.

MVCC-DOC-001 · replit.md + threat_model.md + CHANGELOG.md

Camelot: ADA × TARANTULA | **ZPOS:** SYNTHESIS 37% | **JCSE:** 47/50 | **Deps:** all prior

`replit.md`: project overview, Phase-J-current architecture, score glossary, routes table, user-preferences block (empty). `threat_model.md`: assets, trust boundaries, scan anchors, STRIDE-per-surface, ignore list (Vite dev server, mockup sandbox, tests, attached_assets). `CHANGELOG.md`: single MVP-launch entry summarising the 24 prompts and the seven CLASS-A features. All three files kept under 600 lines apiece; longer detail moved into `.agents/memory/` topic files.

6. Part 4 — Upgrade Path Document

Every CLASS C item below is **triggered** — it ships when the trigger fires, not before. Engineers reading this document should treat CLASS C as a deliberate `// TODO when X` rather than as a missing requirement.

6.1 Triggered upgrades grouped by trigger family

Trigger family A — School / Institution SKU

- Cohorts (instructor / students / CSV) · 4 d
- SAML/OIDC SSO · 5 d
- A11y formal audit (WCAG 2.1 AA) · 5 d
- Department reference + /enterprise heatmap · 3 + 4 d

Trigger family B — Marketplace network effects (≥100 listings · ≥25 creators)

- Roundtable awareness graph · 3 d
- Complementary pairs API · 2 d
- Card synergies cache table + precompute job · 1 + 1 d
- Roundtable state snapshot table · 1 d
- Synthesis sessions (multi-creator) · 5 d
- Endorsements (give + display) · 2 d
- Knight ranks (GUIN+ gamification) · 2 d
- GUIN+ public profile pages + sidebar identity card · 2 + 1 d

Trigger family C — PRO billing live

- Claude narrative on dashboard + resume-narrative button · 1 + 0.5 d
- /report PDF export (jsPDF + html2canvas) · 2 d
- AI usage ledger UI · 1 d
- Subscription cancel + dunning · 2 d
- Billing receipt emails (Resend) · 1 d

Trigger family D — Live Stripe

- Stripe live webhooks (real, not stub) · 4 d

- Public status page · 1 d

Trigger family E — Investor / pre-Series A

- /demo public investor page (Sarah Chen) · 2 d
- A11y formal audit · 5 d
- Datadog RUM dashboards · (already deferred upstream)

Trigger family F — Support / ops load

- Bonsai DAG react-flow visualisation · 3 d
- Bonsai dependency badge per stage · 1 d
- Notifications stream + read state · 3 d
- DRM event ingest + violators · 2 d
- AI cache admin UI · 1 d
- Admin backfill UI · 1 d
- Test results dashboard · 2 d
- Custom CCGE scenario generation · 2 d
- Admin CCGE compendium import · (covered by seed for MVP)

Trigger family G — Locale / mobile

- i18n shell · 4 d
- Mobile-responsive Recharts variants · 3 d

Every triggered upgrade above has an estimate sized for a single developer using Replit Agent with the MVP codebase already in place. Total deferred effort: ~74 developer-days — i.e., one developer can clear the entire backlog in roughly four months *if every trigger fires at once*, which by definition they won't.

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7. Phase K Addendum — Corporate SPHINX Marketplace (Post-MVP Activation)

Status: shipped · feature-flagged OFF by default (`corporateMarketplace`) · graduates from CLASS C on first institutional licence trigger.

What it is. A second SPHINX surface scoped to a single institution. Members of the same `users.institution` (string match, trim + case-insensitive) can browse, publish to, and buy from an internal marketplace that is invisible to the public SPHINX. Listings carry an explicit visibility scope: `OPEN` (public only), `CORPORATE` (institution only), or `BOTH` (dual-publish). The institution string is snapshotted onto the listing at publish time — no foreign key, no cross-tenant join.

Why now. Original PDD deferred SPHINX-Roundtable, complementary pairs, and synthesis under trigger "100+ listings / 25+ creators." Corporate Marketplace is the missing trigger family for the **first institutional licence** (SCHOOL_STUDENT or ENTERPRISE plan) — the moment one buyer wants a private SPC trading floor. It ships at zero cost to the public market: scope filtering is a single predicate added to the existing browse query.

Compression check (SPARTAN gates).

```
NEW SCHEMA       +2 cols (spc_listings.scope · spc_listings.institution)
                 +1 table (spc_feedback · 6 cols · 1 unique index)
NEW ENDPOINTS    +3 (GET corporate/listings · POST + GET
listings/:id/feedback)
NEW CLIENT PAGE  +1 (/marketplace/corporate · reuses ListingDetail for
/:id)
NEW PROMPT       MVCC-MKT-005 · single prompt scope (schema + storage +
3 routes
                 + scope toggle + corporate page + feedback widget +
nav)
NET PROMPT COUNT 24 → 25 (one prompt added · zero existing prompts
modified)
```

MVCC-MKT-005 · Corporate SPHINX Marketplace (post-MVP activation prompt)

Add `SPC_SCOPES` (`OPEN` / `CORPORATE` / `BOTH`) and `SPC_FEEDBACK_BONUS_BY_STARS` (`{1:0, 2:0, 3:2, 4:6, 5:10}`) to `shared/schema.ts` . Add `scope` + `institution` columns to `spc_listings` and a new `spc_feedback` table (listing_id, buyer_id, stars 1-5, comment, bonus_awarded, created_at) with a unique `(listing_id, buyer_id)` index. Idempotent migration `migrations/0007_corporate_marketplace.sql`. Storage adds `getCorporateListings(institution, filters)`, `submitSpcFeedback` (atomic txn: insert feedback + award `SPC_FEEDBACK_BONUS_BY_STARS[stars]` credits to creator + mirror into `listing.totalEarned`), and `getSpcFeedbackForListing` (returns `{count, average, histogram, recent[{buyerName}], mine}` — no raw buyer/creator IDs in the payload). `getAllSpcListings` now filters `scope === "OPEN"` so corporate-only rows can never leak into the public market. Three new routes, all `requireFeature("corporateMarketplace")` + `requireAuth`: `GET /api/sphinx/corporate/listings` (institution-scoped, 403 if caller has no institution), `POST /api/sphinx/listings/:id/feedback` (buyer-only, 409 on duplicate), `GET /api/sphinx/listings/:id/feedback`. Publish route accepts `scope` (defaults `OPEN`), snapshots `creator.institution`, and uses the server-side `isFeatureEnabled` resolver (honors `FEATURE_CORPORATE_MARKETPLACE` env overlay) — never the static shared map. Purchase route enforces same-institution buyer for `CORPORATE` listings (403) before the credit-debit txn. Client: `/marketplace/corporate` page with institution header + pillar filter + listing grid (links to existing `/marketplace/:id`); scope toggle on PublishPage (Open / Corporate / Both, disabled unless flag on + user has institution); `SpcFeedbackPanel` on ListingDetail with star input + comment + bonus-preview ("creator earns N credits") + recent reviews. Nav entry under Explore, route and link both flag-gated.

Threat-model deltas.

```
ELEVATION OF PRIVILEGE   Corporate purchase gate (403 on inst mismatch)
runs
                         BEFORE executePurchase — no wasted credit
debit.
INFO DISCLOSURE          Feedback payload redacts buyerId/creatorId;
resolves
                         buyer display names via batched users lookup.
TAMPERING                Bonus credit award + listing earnings mirror
happen
                         in the SAME txn as feedback insert; unique
                         (listing_id, buyer_id) index is race-safe.
SPOOFING                 Scope is never client-trusted on read — open
browse
                         hard-filters to OPEN; corporate browse derives
                         institution from session, not query string.
```

```
FLAG CONSISTENCY          Server uses isFeatureEnabled() (env-aware), client
                          uses FEATURES map; routes return 404 (not 403) when
                          flag off — indistinguishable from unimplemented.
```

Graduation trigger added to Part 4.

```
Trigger family H — First institutional licence (SCHOOL_STUDENT or ENTERPRISE)
- Corporate SPHINX Marketplace (Phase K) · activated · 0 d (shipped flag-off)
- Set FEATURE_CORPORATE_MARKETPLACE=true at deploy time · 0 d
- (Optional) Per-institution analytics rollup on /enterprise · 2 d
```

Operational notes. The flag is OFF on every deployed environment until the first institutional contract is countersigned; flipping `FEATURE_CORPORATE_MARKETPLACE=true` lights up the surface without a code deploy. Existing OPEN-market behaviour is unchanged: the public list excludes `CORPORATE`-only rows but still surfaces `BOTH`. No data migration is required for tenants that never opt in — the new columns default to `'OPEN'` / `NULL`.

8. Phase L Addendum — Shareable CCGE Badges (Post-MVP Activation)

Status: shipped · public by default · graduates from the social-loop trigger family (zero-cost referral channel for CCGE Arena).

What it is. Every finished CCGE session that crosses the Bronze JCSE threshold (≥30) gets a server-rendered 1200×1200 cyberpunk badge PNG and a public share page with Open Graph + Twitter Card unfurls. The badge displays the player's tier ring (Bronze / Silver / Gold / Platinum), JCSE score, scenario title and tier, awardee name + username, current ARK score, cert level, finish date, and the last-eight of the user's `arkIdString` as a forgery anchor. From the CCGE result screen players get one-click Open · Copy link · Download PNG · native `navigator.share` actions.

Why now. CCGE Arena is the highest-frequency surface in the MVP loop (one-to-many sessions per user per week) and the only one with a natural "win moment" that maps cleanly to a shareable artefact. Originally deferred under the assumption that social proof requires GUIN+ public profiles (CLASS C, gated on 100+ sellers), but the badge surface ships **without** the profile pages: the badge IS the unfurl. Unguessable UUID session IDs make the share link itself the opt-in — no flag, no toggle, no extra schema.

Compression check (SPARTAN gates).

```
NEW SCHEMA       +0 cols · +0 tables (session UUID is the bearer; no
opt-in flag)
NEW ENDPOINTS    +2 (GET /badge/:sessionId.png · GET /badge/:sessionId)
NEW CLIENT UI    +1 component (ShareWinCard on /play result view)
NEW PROMPT       MVCC-CCGE-003 · single prompt scope (Satori renderer +
Resvg PNG
                 + share-page HTML + rate-limit + in-process PNG cache +
UI)
NET PROMPT COUNT 25 → 26 (one prompt added · zero existing prompts
modified)
DEPENDENCIES     +2 npm (satori · @resvg/resvg-js) · 1 outbound fetch
(Inter OTF
                 via jsDelivr v3.19 mirror, cached per process)
```

MVCC-CCGE-003 · Shareable CCGE Badge (Satori renderer + public share page)

`server/badge/render.ts` : Satori JSX-as-plain-objects pipeline rendering a 1200×1200 SVG, converted to PNG via `@resvg/resvg-js` . Layout: header strip (`ARK`

`PLATFORM · CCGE ARENA · CERTIFIED`), centered tier disc with cyan glow (tier label + JCSE numeric + `/50` rule), scenario block (`SCENARIO · <TIER>` + title), footer row (awardee name + `@username`, ARK score + cert level, finish date + `ARK-ID <last8>`). HSL palette matches the project (cyan `188 86% 53%` + dark `222 47% 11%`). Inter Regular + Bold OTFs loaded once from `cdn.jsdelivr.net/gh/rsms/inter@v3.19` and cached for process lifetime.

`server/badge/routes.ts`: two public routes, both gated to `status='finished'` sessions that crossed the Bronze JCSE threshold (returns 404 otherwise — flagged surfaces indistinguishable from missing). `GET /badge/:sessionId.png` returns the PNG with `Cache-Control: public, max-age=3600, s-maxage=86400` + `X-Robots-Tag: noindex`; protected by a per-IP rate limit (30/min via `express-rate-limit`) and an in-process FIFO PNG cache (50 entries, 24h TTL, ~17 MB worst case). Cold render ≈3.6s on first hit (Satori + Resvg + font fetch); cached hit ≈8ms — 450× speedup keeps the surface safe under share-storm traffic. `GET /badge/:sessionId` returns an HTML share page with inline `<style>` and a full Open Graph + Twitter Card meta block: `og:title` = `"<Name> — <Tier> on ARK CCGE"`, `og:image` = same-origin PNG (1200×1200), `og:description` summarises JCSE + scenario + ARK-ID. Page CTA links back to `/play` for the click-through.

`registerBadgeRoutes(app)` mounts in `server/routes.ts` at the top of `registerRoutes()` so the routes attach before the Vite catch-all in dev. `client/src/pages/play.tsx::ShareWinCard` shows on the result view only when `kcseScore >= JCSE_TIER_THRESHOLDS.BRONZE`, with four actions: Open share page (new tab), Copy link (clipboard), Download PNG (signed filename), native Share (uses `navigator.share` where supported, falls back to copy).

Threat-model deltas.

```
INFO DISCLOSURE          Share link is bearer auth via UUID session id —
same
                         model as one-time signed URLs. Badge payload
exposes
                         only name/username/score/scenario/tier/date
already
                         intended for public display. Last-8 of
arkIdString is
                         a hash suffix; no PII or reversible identifier.
DENIAL OF SERVICE        Satori + Resvg are CPU-heavy (~3.6s cold). Per-
IP rate
                         limit (30/min) + FIFO PNG cache (24h TTL) keep
the
```

```
                         endpoint safe under share-storm traffic. Cache
hits
                         drop to ~8ms — no re-render under repeat loads.
SPOOFING                 ARK-ID last-8 printed on badge ties the
artefact to
                         the awardee; forging requires both the session
UUID
                         and the user's arkIdString suffix.
CSP                      Page uses inline <style> and same-origin <img>;
                         covered by existing prod CSP (style-src 'self'
                         'unsafe-inline'; img-src 'self' data:).
ROUTE ORDERING           /badge/* is mounted before the Vite catch-all
in
                         dev and before the static SPA fallback in prod
— no
                         shadowing in either environment.
ROBOT EXPOSURE           X-Robots-Tag: noindex on the PNG keeps share
images
                         out of search indexes; HTML page is
intentionally
                         indexable as a referral surface.
```

Graduation trigger added to Part 4.

```
Trigger family I — Social referral loop (zero-cost growth channel)
- Shareable CCGE badges (Phase L) · activated · 0 d (shipped public)
- Vendor Inter OTFs locally to remove cold-start CDN dependency · 0.5 d
- Optional: per-session isPublicBadge flag + revocable share token,
  if product policy ever requires stronger opt-in than bearer-UUID · 1 d
- Optional: per-tier badge variants (different glow / accent per tier) ·
1 d
```

Operational notes. No env flag — the surface ships ON by default because the bearer-UUID model is the opt-in. There is no DB migration; reads are scoped to existing `game_sessions` + `ccge_scenarios` + `users`. The PNG cache lives in process memory (cleared on every redeploy), which is acceptable for the current single-instance autoscale config; if the deployment ever spreads across multiple instances the cache becomes per-instance (still safe — just lower hit rate). Inter font is fetched on first cold start of each instance; if jsDelivr is unreachable the first render fails with a 500 and the next one retries — vendoring the OTFs locally (the follow-up above) closes that single external dependency.

End of MVP PDD — SPARTAN-compressed from ARK PDD v3 · 26 prompts · 4 weeks + Phase K (Corporate Marketplace) + Phase L (Shareable CCGE Badges) activations · Replit Agent ·

ATANDA Studio · May 2026.